

Buck Boost Regulator (B²R) for Spacecraft Solar Array Power Conversion

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Abstract—Solar Array Regulators (SAR) in a satellite Power Conditioning Unit (PCU) are implemented to transfer the power during sunlight from the solar arrays to the power bus. The objective of this paper is to present how Buck Boost Regulators (B²R) could be used as Step-Up, Direct Energy Transfer and Step-Down Solar Array Regulators. Control circuitries of a new kind are introduced for regulated and unregulated buses, illustrated by practical results demonstrating that such regulators can be used in satellite power systems.

I. INTRODUCTION

Today two different Solar Array Regulator (SAR) techniques are commonly applied on spacecraft platforms. The first family is called Direct Energy Transfer (DET) [1][2][3]. It applies to topologies where the Solar Array Power (PSA) is transferred to the power bus by directly connecting the Solar Array (SA) to the power bus, when power is requested. When no power is requested, the solar array is either shunted or disconnected from the power bus (open circuit), and therefore no solar array power is transferred to the bus. The Sequential Switching Shunt Regulator (S3R) [4][5][6] is probably the most popular DET solar array regulator thanks to its simplicity, good power density and its excellent power in versus power out efficiency when full solar array power is required for the power bus (only diodes and harnesses/connectors conduction losses).

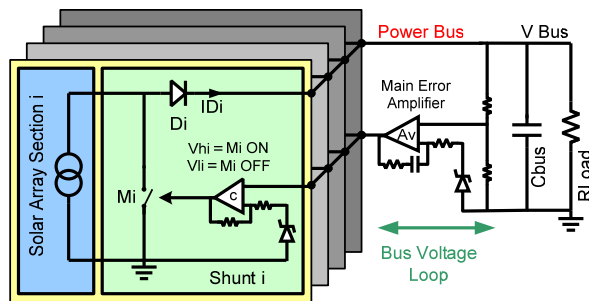


Figure 1. S3R with 4 solar array sections

Unfortunately DET power systems require specific adaptation of the regulator electronics according to the solar array design or vice versa. Moreover when a section is attached to the bus the solar array power delivered to the bus depends on both solar array characteristics and power bus voltage level. This is particularly a negative point for space missions where the solar array characteristics vary widely (due to temperature - as seen in Fig. 2 - , ageing, radiation, solar aspect angle, sun intensity) or when a the battery is directly connected to the power bus which presents some voltage variation.

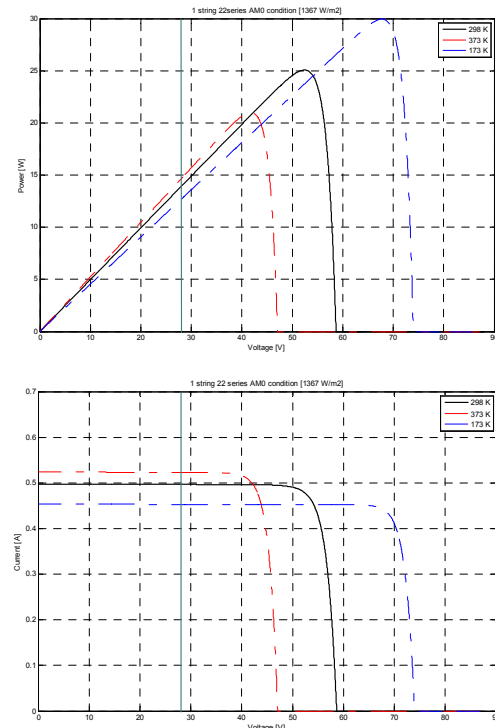


Figure 2. Example of {PSA, VSA} {ISA, VSA} Solar Array characteristics - 1 string of 22 cells in series at 28 degC and +/-100degC (BOL)

As a consequence, for these missions (e.g. deep space missions or Low Earth Orbit missions), DET techniques might be conveniently replaced by Pulse Width Modulation (PWM) switching SAR with Maximum Power Point Tracking (MPPT), capable of extracting the solar array maximum power. To the author's knowledge so far only step-down conversion where the solar array Maximum Power Point (MPP) voltage is higher than the bus voltage (e.g. Globalstar 1 [7], Rosetta, Mars Express, Venus Express [8][9][10], Goc and future missions as Swarm [11], Lisa Pathfinder [12], GAIA), or step-up conversion where the solar array MPP voltage is lower than the bus voltage (e.g. Bepi-Colombo [13][14], Globalstar 2) have been considered for European spacecrafts. The main advantage of both solutions is the fact that the same solar array regulator design can be used for different missions. On the other hand the main drawback of MPPT-SAR is that the switching power converter introduces significant losses (a few percents), which can completely offset the advantage provided by the operation at the maximum power point. Furthermore the efficiency of the SAR drops when the difference between the solar array voltage and the bus voltage increases: the further away the MPP voltage (VMPP) is from the bus voltage, the lower the SAR efficiency is (this is true both for step-up and step-down converters).

In fact the best possible Power Conditioning Unit (PCU) would be one using converters capable to track the maximum power from the solar array in any condition, with an efficiency equal or very close to the efficiency of a power conditioning unit based on DET concept and flexible such that it could be used for different types of missions without any design modifications, reducing thus its cost. The sizing of the system would be defined at the mission conditions with the minimum power margin (e.g. when the available SA power minus the output power demand is minimum) with the SAR operating in DET or close to DET such that the best efficiency is achieved.

II. STEP-UP DET AND STEP-DOWN CONVERSION WITH BUCK BOOST REGULATOR (B²R)

The idea of using a PWM SAR based on step-up and step-down conversion with an MPPT is born from the problematic previously enounced. One has to realize that only a few topologies [15] [16] [17] using two power switching cells have the possibility to operate in step-up, step-down and also in DET conversion mode. Based on this remark the five topologies introduced shown in Fig. 3 have been selected. Their known performances (efficiency), simplicity and frequency responses in continuous mode with a traditional switching frequency (100 KHz) render them suitable for a use in spacecraft power system with either a constant regulated bus voltage or a variable bus voltage with the battery directly connected to the bus. By using such converters the solar array Maximum Power Point Voltage (VMPP) is in all conditions closer to the bus voltage than if simple buck or boost converters are used. Consequently the converter operates always in DET or close to it, improving greatly the efficiency (minimization of the losses).

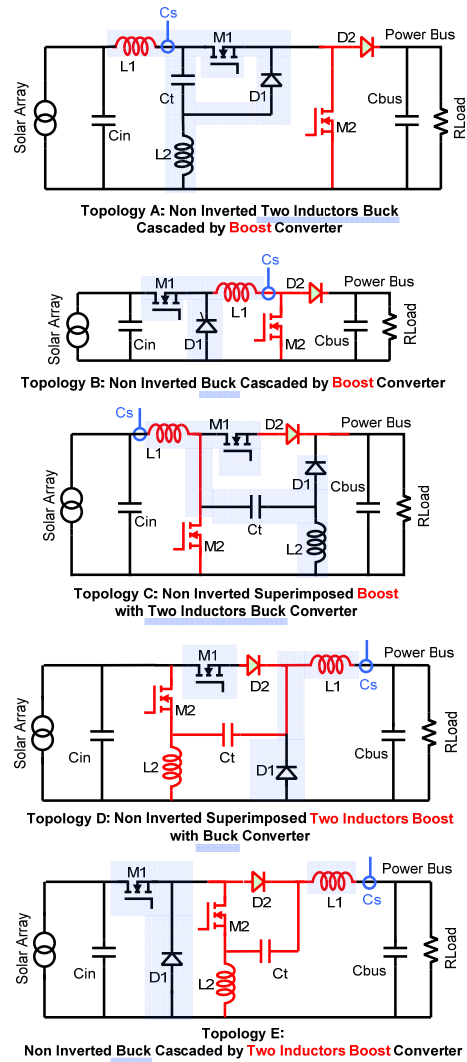


Figure 3. B²R topologies

III. B²R CONTROL LOOPS DESCRIPTION

At the time of the Rosetta development, several control schemes were studied in the European Space Agency Power Lab for the PCDU Step-Down solar array power converter in MPPT mode. The final decision was to implement two inductors buck as SAR with a current loop controlling the output current of the SAR and an input voltage loop embedding the output current loop. The input voltage loop has the main advantage of being stable on the voltage ($V_{SA} > V_{MPP}$) and current region ($V_{SA} < V_{MPP}$) of the solar array characteristics. In case of a power deficit the MPPT becomes active and provides to the input voltage loop a reference signal to let the SAR extract the SA maximum power by oscillating around the MPP. More details are provided concerning the design of this input voltage loop in [8][18][19]. The control of the input voltage provides also some interesting capabilities: indeed for the Rosetta, Mars Express and Venus Express PCUs it is possible by command to set a minimum SA Voltage (VSA) limit with respect to the

relevant solar array $\{I, V\}$ curve, while the MPPT feature can be enabled or disabled. This feature has been successfully used during the Mars Express mission, where it has been possible to recover the spacecraft power despite a wrong inter-connection between the SA and the PCU [10].

Based on this experience it was decided in an early phase of this project to try to implement a similar control scheme as the one used in Rosetta satellite, capable of controlling the input voltage of the step-up DET and step-down topologies of Fig.3 when operating in MPPT mode. Several simulations with average models of the converters [20][21] were performed to find a control scheme respecting the stability requirements applicable to space power equipments (60° Phase and 10dB gain margins) [22][23]. Fig.5 and Fig.6 present two final control schemes for the 5 topologies of Fig. 3 either for battery bus or for a regulated bus spacecraft.

For both architectures two comparators are implemented to generate the buck and boost PWM duty cycle signals. A single current amplifier is connected to the two PWM comparators. Such implementation is an asset to use several B²R in parallel (current sharing) and drive more power and/or for redundancy purposes. As it can be seen on Fig.4 the buck sawtooth is always lower than the boost sawtooth such that the buck and boost cells never switch at the same time (in boost mode the buck switch is always ON, and in buck mode the boost switch is always OFF).

The current loop controls the current in the inductor where the placed on the positive power path. On Fig.3 the current sensing is indicated by the symbol CS. The inductor current loop is then embedded in an input voltage loop, which allows the system to control the input voltage of the SAR in MPPT mode.

It is noted that the topologies A, B, and C contain a boost. It is well known that boost input to output transfer function presents typically a Right Half Plane Zero (RHPZ). The presence of the RHPZ at mid frequencies causes difficulties to achieve high output loop bandwidths. The A, B, and C topologies were therefore primarily studied for unregulated buses where a low battery voltage and current management loop bandwidth (necessary to avoid oscillation with the RHPZ) is not an issue since the battery (low impedance) is directly connected to the power bus.

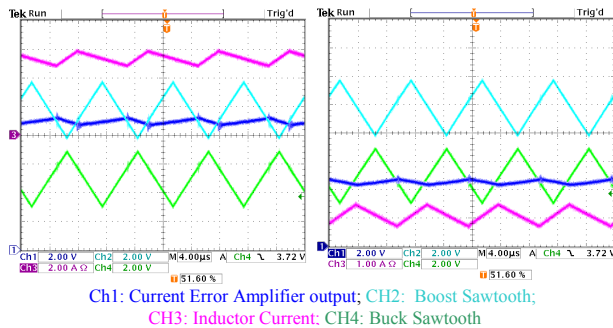


Figure 4. Current Loop waveforms in Buck and Boost mode

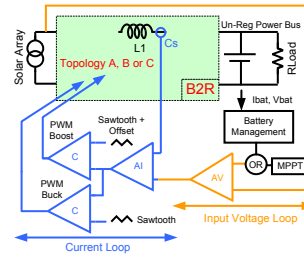


Figure 5. Control scheme for Battery Bus Architecture

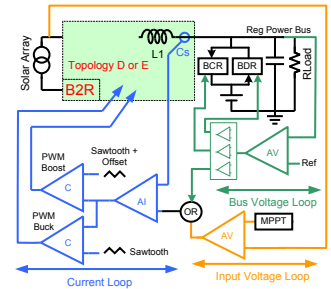


Figure 6. Control scheme for Regulated Bus Architecture

As it can be seen on Fig. 5 the input voltage loop is used either by the battery management (when the available SA power is higher than the battery power bus needs) or by the solar array MPPT (operative in case of SA power deficit).

Topology E or D, containing a two inductors boost, might be more conveniently used for a regulated bus architecture. In excess of SA power an output voltage loop controls the regulated power bus with good performances (no RHPZ thanks to the two inductors boost). In case of SA power deficit, the MPPT controls the input voltage of the converters and the power bus is regulated thanks to the bus voltage loop controlling a battery discharge regulator (BDR).

IV. BREADBOARDS RESULTS

Topologies B and D have been bread-boarded and tested with the control schemes presented on Fig. 5 and Fig.6 (topology B 250W power stage: $C_{in}=49\mu F$, $L_1=50\mu H$, $C_{bus}=22\mu F$ /battery; topology D 450W power stage: $C_{in}=49\mu F$, $L_1=60\mu H$, $C_{bus}=500\mu F$, $C_t=22\mu F/78\mu F+1\Omega$).

For each converter the stability has been studied by measuring the frequency responses of all loops. In particular the Phase Margin (PM), Gain Margin (GM), and Bandwidth (BW) of the input voltage loop were measured in buck mode, boost mode, on the voltage side, current side of the SA characteristics and at the SA maximum Power Point. Fig. 7 and Fig. 8 illustrate the six different operating points at which the open frequency response of the input voltage loop was measured. All measurements confirmed the stability and performances of the input voltage loop in the current and voltage regions of the solar array characteristics, in buck and boost modes for the two breadboards. The results are presented from Fig. 9 to Fig. 20.

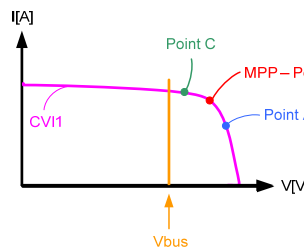


Figure 7. Static Operating points of the SAR in Buck mode when measuring the frequency loop responses

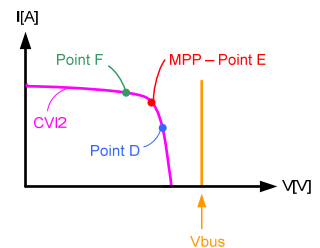


Figure 8. Static Operating points of SAR in Boost mode when measuring the frequency loop responses

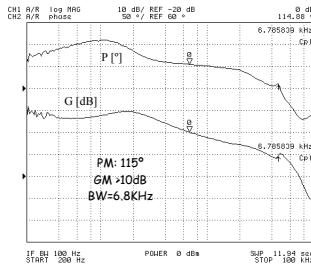


Figure 9. OVL(s) – Buck Mode – Point A – Topology B

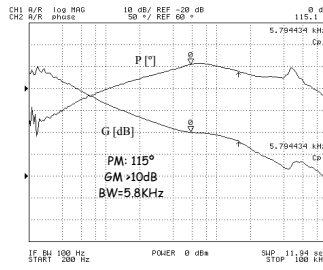


Figure 10. OVL(s) – Buck Mode – Point B – Topology B

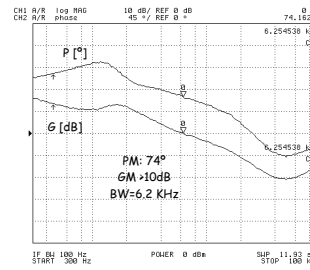


Figure 15. OVL(s) – Boost Mode – Point D – Topology D

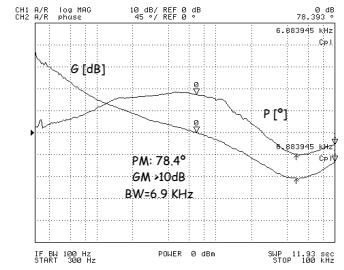


Figure 16. OVL(s) – Boost Mode Point E – Topology D

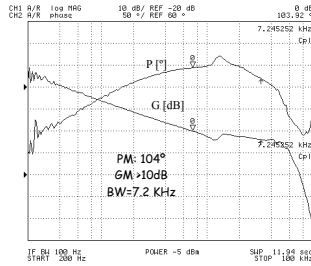


Figure 11. OVL(s) – Buck Mode – Point C – Topology B

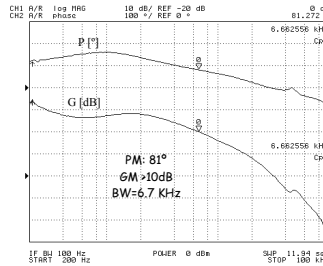


Figure 12. OVL(s) – Boost Mode – Point D – Topology B

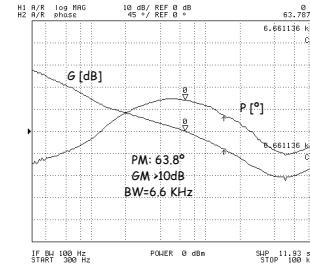


Figure 17. OVL(s) – Boost Mode – Point F – Topology D

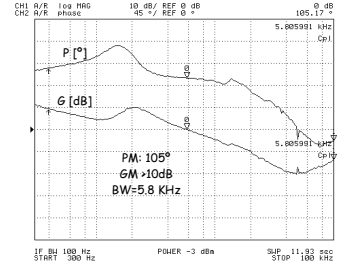


Figure 18. OVL(s) – Buck Mode Point A – Topology D

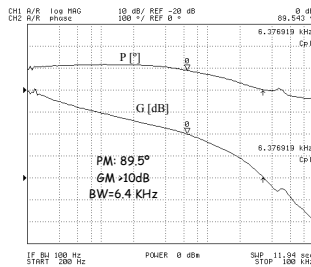


Figure 13. OVL(s) – Boost Mode – Point E – Topology B

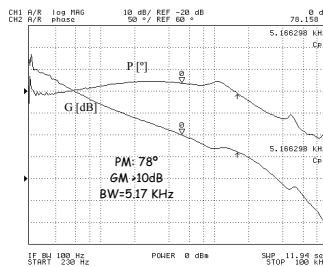


Figure 14. OVL(s) – Boost Mode – Point F – Topology B

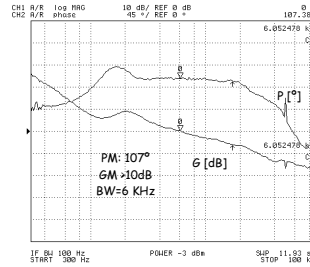


Figure 19. OVL(s) – Buck Mode – Point B – Topology D

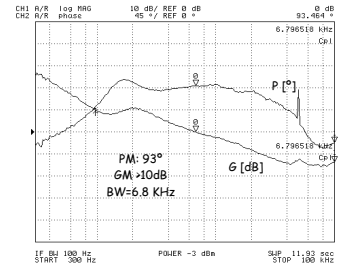


Figure 20. OVL(s) – Buck Mode Point C – Topology D

The input voltage loop performances (bandwidth of several KHz) make this loop compatible for MPPT operation with a traditional tracking frequency of several hundreds of Hz.

Since the particularity of the first B²R breadboard based on the topology B was to use the input voltage loop also for the battery management, the stability of the battery management was measured. The battery management consists in two loops: a current loop in order to charge the battery at a maximum preset current value and a voltage loop reducing the current charge when the battery voltage reaches the preset end of charge voltage. On Fig. 21 and Fig. 22 the Open Battery Current Loop (OBCL) frequency responses are presented when the SAR is in buck and in boost mode. It can be clearly seen on Fig. 22 the effect of the RHPZ in boost mode where the phase goes down and the gain goes up above 1KHz. Due to this effect the bandwidth of the battery current loop has been voluntary reduced to 430Hz to avoid any possible oscillation and meet the stability requirements [22].

Following the frequency responses measurements the B²R breadboards were tested in time domain: as can be seen on Fig. 23, Fig.24 and Fig.25 the B²R is capable to operate in step-up, DET and step-down modes. These results were identical for both breadboards (B and D topologies). VSA represents the solar array voltage, VBUS is the power bus voltage (B²R output voltage) and PSA is the solar array power extracted by the B²R.

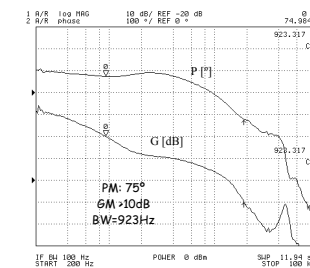


Figure 21. OBCL(s) in buck mode – Point A – Topology B

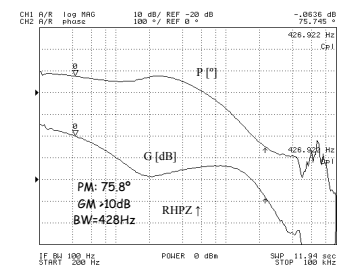


Figure 22. OBCL(s) in boost mode – Point D – Topology B

For the three figures, the B²R operates within the voltage and current regions of the SA: it oscillates around the maximum power point. On Fig. 25 and Fig. 26 VMPP is equal to the bus voltage. The internal waveforms at the PWM comparators level are plotted on Fig. 26, when the B²R has to operate alternately in buck DET and boost modes. One can see the change of operating domain of the current amplifier output between these three modes.

Finally the efficiencies of the two B²R breadboards have been measured for:

- a 100 KHz switching frequency
- an output power from 20W to 250W,
- a minimum and maximum solar array voltages corresponding to the MPP voltages of a solar panel composed of sections with 15 cells in series at +100 °C and -100 °C (26V - 45V)
- a minimum and maximum output voltage corresponding to the classical values of a battery bus variation in nominal operation (34V - 26V)

On Fig. 27 and Fig. 28 the breadboards efficiency results including the driver losses are reported:

- with the B²R in DET mode for the two extreme bus voltages
- with the B²R in buck mode with an input SA voltage of 45V and a bus voltage of 26V
- with the B²R in boost mode with an input SA voltage of 26V and a bus voltage of 34V

The efficiency is greater than 96% including the worst case for the buck and boost modes with the highest difference between the MPP SA voltage and the battery bus voltage. Please note that no protection was present on the B²R breadboards, which would increase the losses and reduce the efficiency in a real application.

V. B²R APPLICATIONS

One aspect that has not been treated until now is the use of B²R in a Single Point Failure Free (SPFF) spacecraft power system. A Single point failure free design is characterized by the fact that no degradation is foreseen after a single failure.

Fig. 29 illustrates the case of a battery bus architecture using as SAR the topology B of Fig.3 in a satellite having two SA wings with different characteristics. On this figure four SARs are used: two SARs are in hot redundancy for each wing such that if one fails, it is disconnected thanks to the protection P1. One can see that an extra diode (D3) has been placed in series with the boost diode D2 in order to protect the bus in case of a short circuit failure of the diode D2.

Dedicated MPPT and Input Voltage Loop (IVLA and IVLB) regulation are implemented for each wing. These loops are triple majority voted with majority voters (MV) in order to guarantee the input voltage control even after a failure.

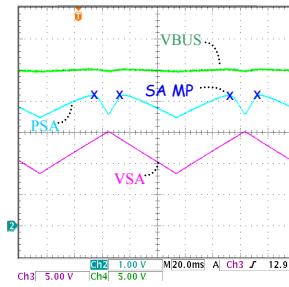


Figure 23. Input Voltage Modulation, with the SA MP Voltage below the Bus voltage

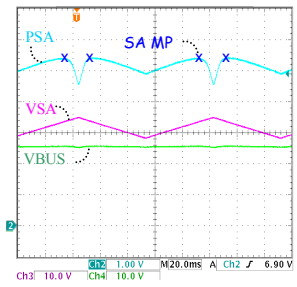


Figure 24. Input Voltage Modulation, with the SA MP Voltage above the Bus voltage

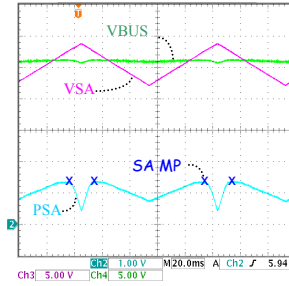


Figure 25. Input Voltage Modulation, with the SA MP Voltage equal to the Bus voltage

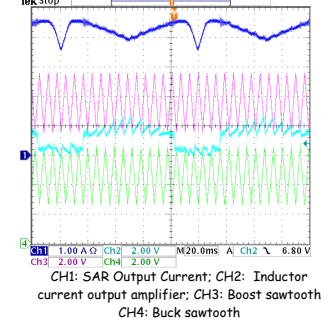


Figure 26. Input Voltage Modulation, with the SA MP Voltage equal to the Bus voltage

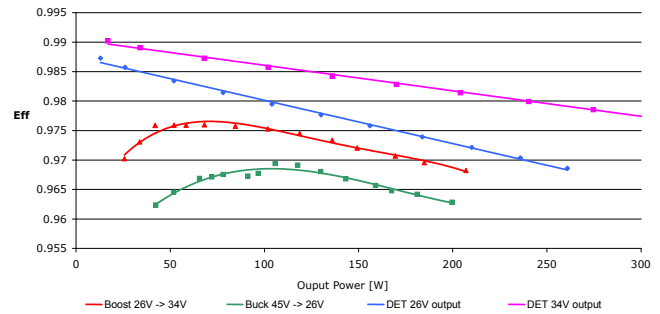


Figure 27. Efficiency measurements of the B²R breadboard with the topology B

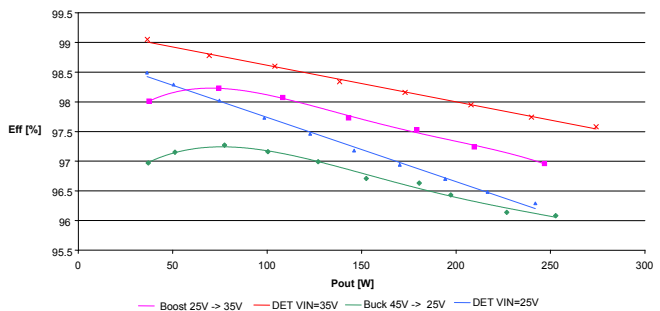


Figure 28. Efficiency measurements of the B²R breadboard with the topology D

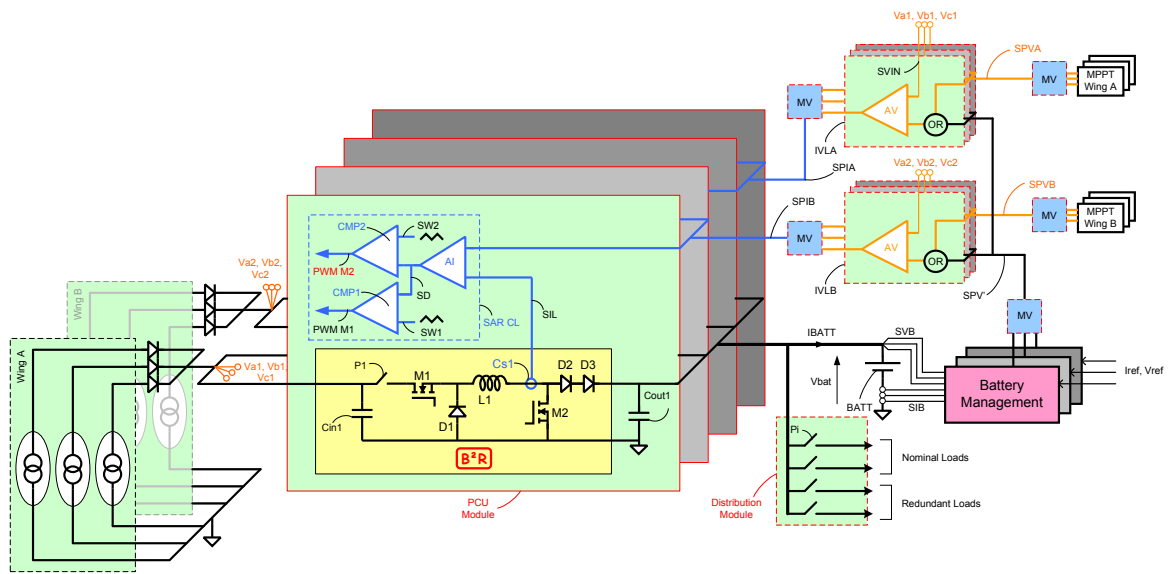


Figure 29. B²R in a Single Point Failure Free Battery Bus Power Architecture

The battery management (also triple majority voted for reliability reason) controls the Input Voltage Loop (IVLA and IVLB) when the spacecraft is in excess of SA power. Note that in this condition the operating point of the SARs is in the voltage region of the solar array characteristics. In case of a SA power deficit, the MPPT associated to the wing starts to operate and oscillates around the MPP by controlling the input voltage of the SAR.

It may be clearly seen that the system has the capability of having one group of SAR connected to the first wing in MPPT mode and the other group of SAR connected to second wing in Battery management mode when the two spacecraft solar array wings have different MPP.

Fig. 30 illustrates the case of a regulated bus architecture by using topology D of Fig. 3 as SAR. On this figure the satellite has also two SA wings with different characteristics. Two SARs are in parallel for each wing. Should one SAR fail, it would be disconnected thanks to the protection switches P1 and PGND.

In case the SA power excess, the SARs regulate the main power bus thanks to the triple majority voted Output Voltage Loop (OVL) controlling the Solar Array Regulator Current Loop (SARCL) with the MEA signal. Note that in this condition the operating point of the SARs is in the voltage region of the solar array characteristics.

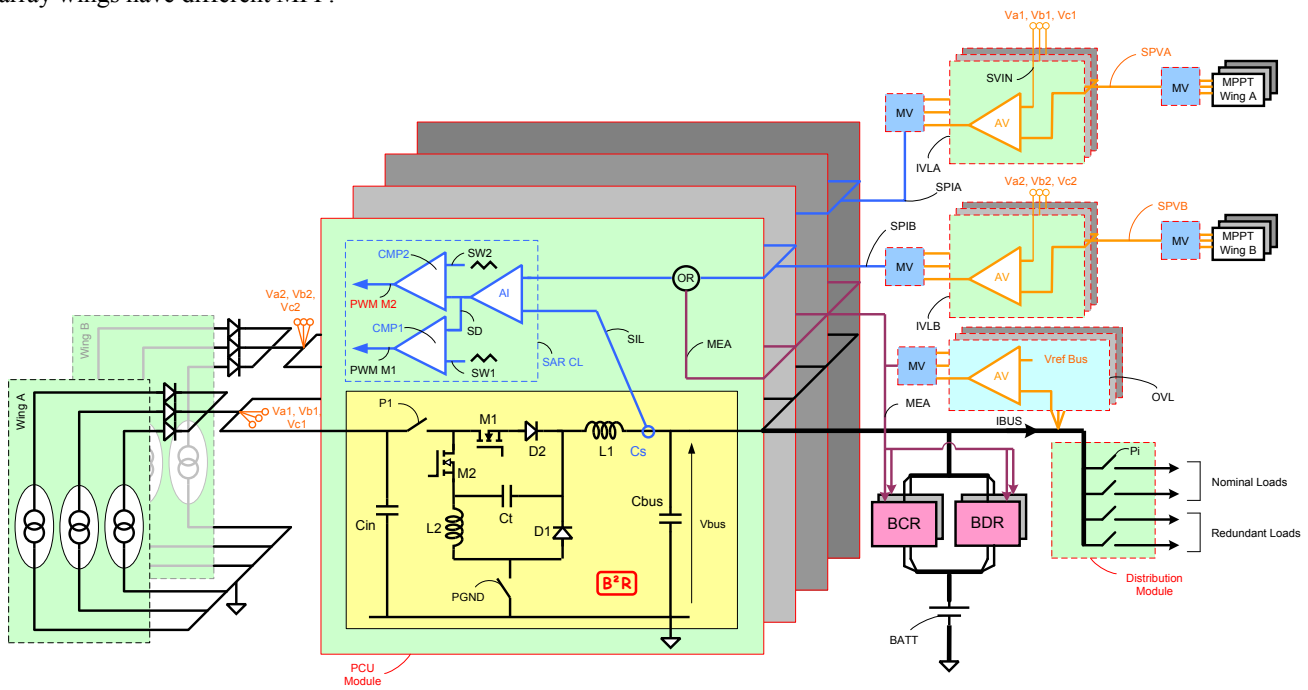


Figure 30. B²R in a Single Point Failure Free Regulated Bus Power Architecture

When SA power is in excess, the SARs regulate the main power bus thanks to the triple majority voted OVL controlling the SARCL with the MEA signal. Note that in this condition the operating point of the SARs is in the voltage region of the solar array characteristics.

In case of a power deficit, the MPPT associated to the solar array wing starts to operate and controls the input voltage of the SAR. If both wings are in MPPT mode, the bus regulation is guaranteed by the OVL controlling the Battery Discharge Regulator (BDR) with the MEA signal.

Finally another interesting property of using B²R is that for both architectures it is possible to characterize the solar arrays during Assembly Integration and Test (AIT) and during flight. As it can be seen on Fig. 31 and Fig. 32, by directly accessing the control circuitry of the step-up, DET and step down SAR breadboards, the solar arrays have been characterized from 0V up to their Open Circuit Voltage (VOC). Such a feature might be an interesting option in order to check the health of the spacecraft solar arrays and to learn more about their characteristic evolutions through a mission.

VI. CONCLUSION

Buck-Boost Regulators (B²R) with a simple control of a new kind have been studied and tested. Practical measurements have confirmed the good efficiency and performances of the proposed control scheme rendering B²R suitable for solar array maximum power point tracking. Two single point failure free satellite power architectures have been introduced to the reader to illustrate how B²R can be used in a space application.

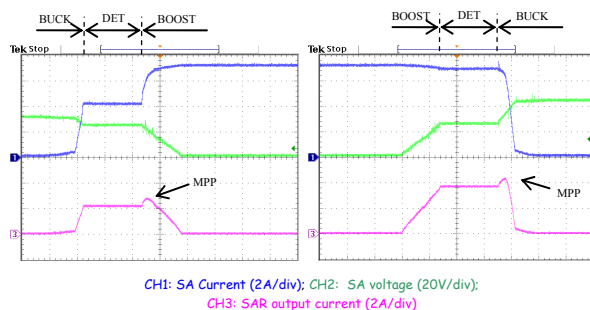


Figure 31. Characterisation of the SA with the MPP Voltage below the Bus voltage

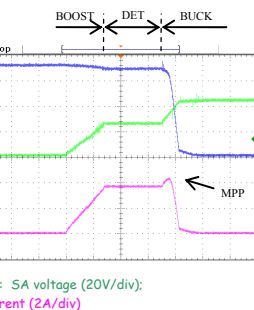


Figure 32. Characterisation of the SA with the MPP Voltage above the Bus voltage

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